

# Slime blocks moved by pistons often fail to bounce up the player

MC - 89043 Redstone

|          |                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------|
| Status   | Resolved / Fixed                                                                                                      |
| Affected | Minecraft 15w38b, Minecraft 15w39b, Minecraft 15w39c, Minecraft 15w41b, Minecraft 15w42a, Minecraft 15w45a (+92 more) |
| Fixed in | 20w07a                                                                                                                |
| Reported | 2015-09-18                                                                                                            |

## STEPS TO REPRODUCE

1. Enter a Minecraft world.
2. Place a sticky piston facing upward.
3. Place a slime block on top of the sticky piston.
4. Place a redstone block beside the slime block.
5. Update the piston so that the slime block moves up and down.
6. Jump onto the moving slime block and observe that the player is only rarely bounced upward.

## OBSERVED BEHAVIOUR

## Actual behavior

When the player lands on the slime block while it is being moved vertically by the sticky piston, the player is only launched upward intermittently. Most impacts do not produce the expected bounce, even though the player appears to contact the slime block.

Other entities, such as mobs, are consistently launched by the same moving slime block in the same setup. This indicates that the issue is specific to player movement or synchronization rather than the piston or slime-block movement itself. In particular, the player's upward velocity may not be reliably synchronized to the client after the bounce, causing the launch to be missing or barely visible.

## EXPECTED BEHAVIOUR

When the player lands on a slime block being moved by a piston, the player should be launched upward consistently, rather than bouncing only occasionally. The bounce should occur reliably whether the player jumps onto or falls onto the moving slime block, with the player's upward movement visible and synchronized correctly. Other entities should continue to bounce as expected.

## ENVIRONMENT

Minecraft Java Edition Versions: 15w38b, 15w39b, 15w39c, 15w41b, 15w42a, 15w45a, 15w46a, 15w47a, 15w47b, 15w47c, 15w49a, 15w49b, 15w50a, 1.9 Pre-Release 2, 1.9 Pre-Release 3, 1.9 Pre-Release 4, 1.9, 1.9.1 Pre-Release 1, 1.9.1 Pre-Release 2, 1.9.1 Pre-Release 3, 1.9.1, 1.9.2, 16w14a, 16w15a, 16w15b, 1.9.3 Pre-Release 1, 1.9.3 Pre-Release 2, 1.9.3 Pre-Release 3, 1.9.3, 1.9.4, 16w20a, 16w21a, 16w21b, 1.10 Pre-Release 1, 1.10 Pre-Release 2, 1.10, 1.10.2, 16w43a, 16w44a, 1.11 Pre-Release 1, 1.11, 16w50a, 1.11.1, 1.11.2, 17w06a, 1.12, 1.12.1 Pre-Release 1, 1.12.1, 1.12.2 Pre-Release 1, 1.12.2 Pre-Release 2, 1.12.2, 17w43a, 17w43b, 17w45a, 17w45b, 18w20b, 1.13-pre1, 1.13-pre2, 1.13-pre3, 1.13-pre4, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13, 1.13.1, 1.13.2, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w03c, 19w04a, 19w04b, 19w05a, 19w06a, 19w07a, 19w08b, 1.14 Pre-Release 2, 1.14 Pre-Release 3, 1.14, 1.14.2 Pre-Release 2, 1.14.4, 19w37a, 1.15 Pre-release 1, 1.15 Pre-Release 2, 1.15 Pre-release 3, 1.15 Pre-release 4, 1.15 Pre-release 5, 1.15 Pre-release 6, 1.15 Pre-release 7, 1.15, 1.15.1, 1.15.1 Pre-release 1, 1.15.2 Pre-Release 1, 1.15.2 Pre-release 2, 1.15.2, 20w06a (includes snapshots)

## ORIGINAL DESCRIPTION

## The bug

When a piston moves a slime block players are not always launched.

## How to reproduce

- Place a upwards facing sticky piston

- Place a slime block on top
  - Place a redstone block on the side of the slime block
  - Update the piston
  - Jump on the slime block (which now moves up and down)
- You will see that you just rarely get bounced up. For entities other than the player it works fine.

### ## Example

This video shows it: <https://www.youtube.com/watch?v=qgFt5L5f9jY>

### ## Additional information

It works fine for other entities, so it probably has to do with the client/server communication or similar.

### ## Potential Fix

I noticed there is a flag called "velocityChanged" in MCP for entities. It is for example used when an entity is hit and needs to get knock back.

When the flag is set, the EntityTracker will send an extra packet containing the entities velocity.

Setting this flag when an entity is bounced by a slime block pretty much fixes this issue. (There is still an related one that if you fall on a moving slime block you don't get bounced which might have similar effects.)

Video with a comparison flag set vs flag not set: <https://www.youtube.com/watch?v=VKHBJxtSlc0&t=2m42s>

## MANUAL RESULT

|                  |                    |
|------------------|--------------------|
| Version used     |                    |
| Reproduced?      | YES / NO / PARTIAL |
| Crash file       |                    |
| Notes            |                    |
|                  |                    |
| Tested by / date |                    |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-89043>